Detection performance prediction on IR images assisted by evolutionary learning

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ABSTRACT

Background clutter characterization in IR imagery has become an actively researched field and several clutter models have been reported. These models attempt to evaluate the target detection/recognition probabilities that are characteristic to a certain scene when specific target and human visual perception features are known. The prior knowledge assumed and required by these models is a severe limitation. Furthermore, the attempt to model subjective and intricate mechanisms such as human perception with simple mathematical formulae is controversial. In this paper, we introduce the idea of adaptive models that are dynamically derived from a set of examples by a supervised evolutionary learning scheme. A set of characteristic scene and target features with a demonstrated influence on the human visual perception mechanism is first extracted from the original images. Then, the correlation between these features and the results obtained by visual observer tests on the same set of images are captured into a model by the learning scheme. The effectiveness of the adaptive modeling principle is discussed in the final part of the paper.

1. INTRODUCTION

A wide variety of models have been developed in the past decades that were aimed at predicting the probability of detecting or recognizing military targets in a given scene characterized by clutter[1-5]. Most of these models incorporated psychological and physiological findings about the modus operandi[6,7] and the acuity[8,9] of the human visual search mechanism. Although some of the classic models have achieved a large popularity, a series of recent publications argued about the necessity to modify or update them in order to compensate for some observed limitations[10,11]. In our opinion, the limitations of these models could be partly due to the fact that they attempt to capture a diversity of distinct factors using simple and general formulae. Even modeling the visual acuity of a single observer is an intricate task, since perceptive abilities of individuals are strongly dependent on subtle and subjective factors. Another common characteristic of the models in discussion is that they assume prior knowledge about the targets and, therefore, cannot be automated. Finally, it has been pointed out that significant differences exist between an imager functioning in the visible spectrum and one in infrared[12]. Consequently, the criteria employed for predicting performance in the visible domain, such as Johnson's criteria[13], may not be suitable for infrared. Several researchers have observed this contradiction and have focused their efforts on performance prediction specific to infrared imagery[14,15].

Our conjecture is that a specific model, rather than a general one, may be a solution to some of these limitations. In other words, different models should be generated for distinct situations, observers, types of imaging equipment, etc. However, in this case, due to the specificity of the model, most of the significant factors involved will be almost impossible to predict analytically. The alternative is to let a heuristic technique learn from examples and generate a model specific to the context under which the examples were generated. This latter approach presents several advantages, given that an appropriate learning scheme is employed. Among these advantages, a very important one is that the necessity of prior knowledge about targets and factors involved is much diminished. Furthermore, this method would be able to generate performance models applicable to both human observers and seekers.

Evidently, the learning scheme outlined above must be based on a supervised learning architecture. In the training data file, one must include an objective value that represents the measured performance of the observer/seeker on a set of images. In correspondence with the objective value will be a list of features considered influential for this performance, such as target contrast and extent, or various clutter metrics from the literature[3-5,14]. In this context, the goal of the modeling system will be to extract the correlations that exist between the observed performance and the features presented and capture them into an analytical model.

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In order to accomplish the task defined above, the learning method must meet two criteria. First, the models must be issued in a closed, analytical form, so that they can be interpreted. This criterion disqualifies the neural networks, which have been traditionally employed in predicting the behavior of highly complex processes, as potential candidates for the learning method. Secondly, the proposed method must be able to automatically select the features that are actually relevant from the provided set of features.

2. EVOLUTIONARY LEARNING

As the principal component of the adaptive modeling system described in the introductory part, we have formulated an approach based on evolutionary programming[16] (EP). EP has emerged in recent years as a powerful tool in search, optimization, and classification applications. Essentially, EP is a paradigm based on principles advocated by the disciples of natural evolution. EP is concerned with developing simple computer programs (or generalized functions) that produce a transfer function that models the underlying relations within a data set. The processes that they model can be very complex and non-linear. Accordingly, EP tools can be very effective in modeling or system identification due to their ability to learn from examples.

The physical mechanism of an EP method consists of a few standard steps:

Step 1: Map the solution space of the problem at hand into a genetic space where each possible solution has a correspondent in a string of numbers called an *individual*, or *entity*. In our case, the solutions, or entities, are simple functions that model a data set.

Step 2: Define a population of entities and randomly initialize it.

Step 3: Evaluate the fitness of each individual in the population. The fitness describes the goodness of the solution coded into the individual. In our case, this fitness is the error between the actual and the predicted values. The goal of the evolution is to minimize this error.

Step 4: With probabilities proportional to their fitness (or inversely proportional, if the fitness is to be minimized), pairs of entities are selected for recombination. An operator called crossover that is the algorithmic equivalent of sexual reproduction realizes the recombination. Crossover produces children that compose the entities of the next generation. By selecting parents according to their fitness, it is assured that better solutions propagate their characteristics into the next generations.

Step 5: Apply mutation and other operators. These operators enhance the performance and robustness of the process.

Step 6: If the overall stopping conditions are not met, repeat Steps 3-6 for the next generation.

An example of a practical application solved via EP is depicted in Figure 1. In essence, one attempts to find a model that describes the data set illustrated in the left-hand side of the picture. In the data set are represented the areas and the respective radii for six circles. In this context, finding a model for this data set reduces to finding the formula for the area of a circle of a given radius (`area` $= \pi \times$ `radius`^{2}).

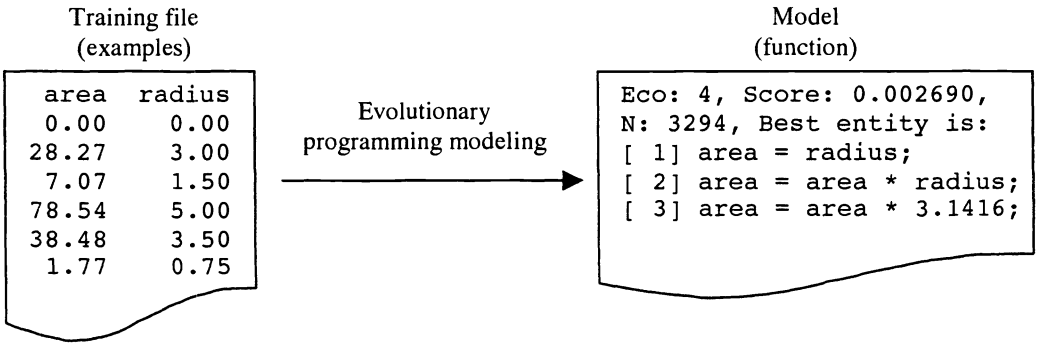

Figure 1. Illustration of a practical application solved by the evolutionary programming package e^{TM}.

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In the right-hand side of Figure 1 is illustrated one of the best models obtained by the EP-based learning method employed (called e^{TM} and developed by System Dynamics International, Inc.). This model represents a translation of the well-known formula into a computer program written in a generic language. Even the number π has been rediscovered with a precision of four decimals. Obviously, the application described above is trivial and has been employed for illustrative purposes only. The modeling power of the proposed method is better demonstrated by the multitude of complex applications successfully addressed via EP in recent years at I-Math Associates.

3. FEATURE SELECTION

An important condition for the success of the model generation process is the quality of the selected features. For our problem, these features must be clearly relevant for the detection process in a wide variety of case conditions. Essentially, they have to capture factors with demonstrated impact on human vision, such as contrast between target and its local background, internal target structure, area and shape of the target, spectral composition, number and density of confusing forms within the scene, etc. A long series of clutter metrics have been reported in the literature in the past two decades[3-5,14] and some of these features can be readily employed by the modeling system proposed in this paper.

It is our conjecture that another category of features may be obtained through image transforms that exploit and, therefore, can quantify descriptive elements such as roughness, self-similarity, repetitiveness, or correlation between neighboring pixels. For example, the Gabor transform achieved a large popularity after several studies showed evidence about the relation between its localization mechanism and the mammalian early vision system[17]. Several candidate transforms for feature generation are currently under investigation. Among these, the wavelet and Gabor transforms, the Markov Random Field models, the graylevel co-occurrence matrices, and the fractal processes have produced promising preliminary results.

An important aspect of the feature selection process has been neglected in the previous paragraphs. All the features mentioned above are global, calculated over an entire scene. Furthermore, the classical metrics referenced in the previous paragraphs produce a single number for an entire image. This number describes the amount of clutter present in the scene and can be considered a detection probability indicator. However, for scenes that contain very small targets, the significance of the global metrics will be severely limited. Indeed, no matter how obvious the small targets may be, they will not affect the global features to an observable extent due to their limited influence on the statistics at the scene level. The answer to this dilemma relies on a multiresolution approach that considers two distinct scales. The features extracted at these two scales are called global and local features, respectively.

3.1. Global features

The list of global features implemented into our adaptive clutter modeling system is shown in Table 1. Also indicated in Table 1 are the formulae, the references corresponding to the different features, and the symbol descriptions. Additional explanations are provided below:

- *The RMS clutter*[3] due to Schmieder and Weathersby is a classical clutter evaluation metric. The image is first partitioned into adjacent windows whose areas are roughly twice as large as the ones of the expected targets. The final metric is derived from the individual metrics calculated over these windows.
- *The clutter invariant*[14] measures the relative difference between the standard deviation of the entire scene and the RMS clutter. It is argued that this normalized difference is due entirely to structure in the image and is background specific.
- *The probability of edge* is the standard deviation of the probabilities of edges calculated over adjacent partition windows. The probability of edge is defined as the probability of a pixel being an edge pixel. The image is first edge-enhanced and thresholded to produce a binary edge map. The recommended edge detector for this operation is the DOOG[18] filter (difference of offset Gaussians).
- *The Kolmogorov-Smirnov test* measures the difference between the actual edge pixel distribution on a scene and a uniform distribution. It is argued that a uniform distribution of the edge pixels corresponds to the highest clutter in the scene. First, the edge map of the image is generated and partitioned into adjacent windows. A cumulative distribution function (CDF) is calculated for each window based on the intensity histogram. The feature value is given by the sum of the absolute differences between the samples of the CDF and a ramp, which is the CDF for the uniform distribution.

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- *The global target prominence* metrics are target-dependent metrics that measure the degree in which the target can be discriminated from its local background. Different prominence measures can be used; in this implementation, we have selected the contrast and the edge-content based measures. The algorithms for calculating the global target prominence features are discussed next.

Metric	Expression	Notations
RMS Clutter[3]	$RMS = \sqrt{\frac{1}{N}\sum_i \sigma_i^2}$	• N: the total number of adjacent square windows considered in the partition of the image; • σ_i^2: the variance of the graylevel distribution of window i.
Clutter invariant[14]	$I = \frac{\sigma - RMS}{\sigma}$	• σ : The global variance of the scene.
Probability of edge[4]	$POE = \sqrt{\frac{1}{N} POE_i^2}$	• POE_i: the probability of edge calculated over the partition window i.
Kolmogorov-Smirnov test[4]	$TC = \frac{1}{L}\sum_{i=0}^{L-1}\left\lvert S_N(i) - \frac{i}{L}\right\rvert$	• $S_N(i) = \left(\sum_{j=0}^{i} N_j\right) \Big/ M^2$: the empirical cumulative distribution function (CDF) over the window; • $\{N_i\}_{i=0}^{L-1}$: the L-level intensity histogram of the window; • M^2: number of pixels of the window.
Global target prominence for contrast[22]	*GTPC* (see algorithm described in text)	
Global target prominence for target edge strength[22]	*GTPE* (see algorithm described in text)	

Table 1. Global features implemented into the modeling system.

GTPC uses a double-window filter transform of the image and is calculated with respect to a certain target present in the scene. For that target, a target region and a background region are created; the target region is the minimum rectangle that encompasses the target, whereas the background region has an area twice as large as the target region and the same aspect ratio. A so-called realization value is calculated next by subtracting the average intensity of the background region from that of the target region and dividing the result by the standard deviation of the background. Then, a transformed image is obtained as follows: for each pixel in the image, two concentric windows equal to the target region and the background region, respectively, are constructed. The difference in average intensity between the target window and the background window is calculated and divided by the standard deviation of the background window. On the normalized histogram of the transformed image, the realization value is marked. Finally, the GTPC is defined as the area under the histogram of the transformed image for values less than the realization value. The GTPC can be referred to as a target dependent signal-to-noise ratio. The difference between the target mean and background mean is a measure of target signal. On the other hand, the standard deviation of the background can be considered a measure of noise around target.

GTPE is similar in principle with GTPC and attempts to evaluate the target prominence based on its edge content. First, a Sobel edge map of the image is derived and transformed by a smoothing operator. The dimensions of the window are equal with the target region. A realization value is calculated for the position where the window coincides with the actual target.

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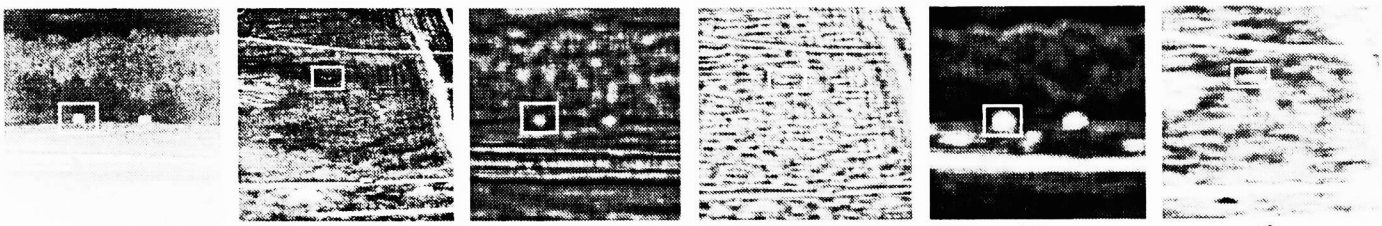
The area under the histogram for values less than the realization value gives the GTPE. This metric quantifies the relative edge information contained in the target region as compared to the entire image.

For illustrative purposes, the two images with different clutter contents shown in Figures 2 (a) and (b) were processed through the global target prominence algorithms described above. The reference targets are encapsulated in a rectangular box. The corresponding transformation maps are shown in Figures 2 (c), (d), (e), and (f), respectively, where the responsiveness of the features to the selected targets is evident.



(a) (b) (c) (d) (e) (f)

Figure 2. Transformation maps obtained by global target prominence features for the original images from (a) and (b): GTPC - (c) and (e), GTPE - (d) and (f).

3.2. Local features

Table 2 lists the local features implemented into our adaptive modeling system:

Metric	Expression	Notations
Target-background contrast[22]	$TBC = \lvert \mu_T - \mu_B \rvert$	• $\mu_{T,B}$ average intensity of target or background, respectively;
Target interference ratio[22]	$TIR = \dfrac{\mu_T - \mu_B}{\sigma_B}$	• σ_B: variance of the local background
Target-background interference ratio[22]	$TBIR = \dfrac{\mu_T - \mu_B}{\sqrt{\sigma_B \sigma_T}}$	
Target-background interference (new)[22]	$TBIR_{new} = \dfrac{\mu_T - \mu_B}{\sigma_B + \sigma_T}$	
Target-background entropy*	$TBE = E(T) - E(B)$	• $E(P) = -\sum_i p(i)\log_2 p(i)$ is the entropy of region P. • $p(i)$ is the magnitude of the ith bin of the intensity histogram (in probabilities).

Table 2. Local features implemented in the proposed clutter modeling system.

Table 2 is self-explanatory. All the local features are calculated over two regions, namely the target region and the background region, as defined in the previous section.

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4. CONFUSING FORM DETECTION AND SEGMENTATION

It can be inferred from our discussion of feature selection that the local clutter modeling methodology requires prior knowledge about the locations and the extent of the target-like objects encountered in the scenes. A wide variety of target detection procedures in visible and infrared images have been proposed in the literature and a comprehensive enumeration of them would be a very ambitious task. As opposed to the various target detection studies, we are concerned with detecting and segmenting all the objects in a scene that may be considered a target by a human observer. This category of objects includes but is not limited to the real targets.

We have implemented and tested several algorithms for confusing form detection and segmentation; the ones that achieved the most promising results are listed in Table 3.

Algorithm	Block diagram	Results
1.	→ Edge detection → Rule-based processing → Contour aggregation →	• Too many false alarms; • Computationally expensive; • Intensive and hard-to-tune rule-based processing.
2.	→ Hit-Miss transform → Scale-space smoothing → Region growing → Detection: Hit-Miss transform; Segmentation: Scale-space smoothing, Region growing	• Computationally expensive (the Hit-Miss transform); • Seeded region growing works poorly on objects with poor contrast.
3.	→ Window tracker → Scale-space smoothing → Region growing → Detection: Window tracker; Segmentation: Scale-space smoothing, Region growing	• Seeded region growing works poorly on objects with poor contrast.

Table 3. Confusing form detector/segmenter algorithms tested.

The various blocks of the algorithms are described in the following paragraphs:

- *The edge detection* module from the first algorithm is based on a DOOG filter.
- *The rule based processing* and *the contour aggregations* modules from algorithm 1 are designed in combination to detect and restore closed contours of target-like objects in the edge map.
- *The hit-miss transform*[19] is a well-known technique to detect objects and generic shapes in clutter. The intensity image is thresholded at N levels to produce N binary maps. A hit (miss) correlation is performed on each of these binary images and the output correlations are thresholded. An AND operation is then applied to each pair of hit/miss correlations and the final N results are fused together. The hit (H) filter is a white rectangle that fits inside the smallest object, whereas the miss (M) filter has a black central rectangle equal to the largest object size. The H filter detects all regions that are equal to or larger than the smallest object. The M filter is applied to the complement image and locates regions that are equal to or less than the largest object. The intersection (AND operation) between the H and M outputs will have peaks at the locations of objects with sizes between the smallest and the largest. This technique detects objects by using both the object shape and its background. A great advantage of the HMT transform is that it allows for a real-time, optical[20] implementation.
- *Scale-space smoothing* is a multiresolution approach to low-pass filtering in which coarser images are generated by convolving the original images with Gaussian kernels of increased variances. The scale-space smoothing takes into

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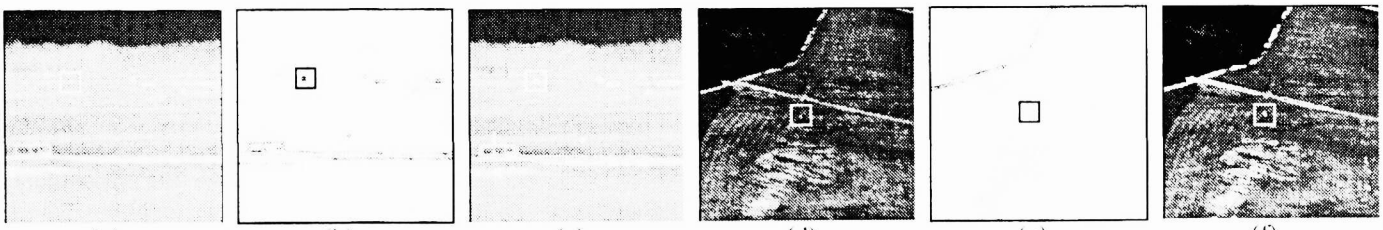
account the edge content of the image at different resolutions. The result is an operator that equalizes the regions with similar graylevels while enhancing the edge configurations.

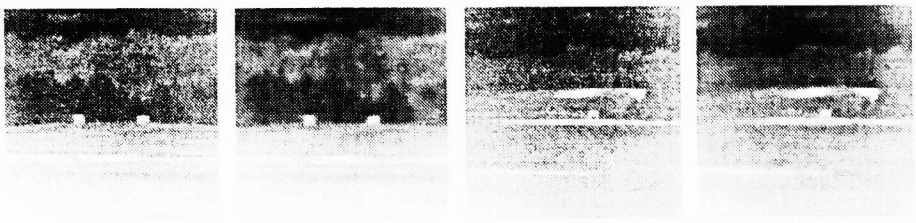
- *The region growing* module is an implementation of a segmentation method based on seeded region growing[21]. Essentially, two regions, the target and its local background, are grown from initial seeds based on pixel adjacency and intensity similarity.
- *The window tracker* is a classical detection filter based on two concentric boxes. Somewhat related to the hit-miss transform, the window tracker requires prior information about the expected target dimensions. The objects are detected as local maxima in a map. This map is generated by representing, for every position of the concentric boxes, the difference between the average intensity within the inner box and the average intensity of the region between the borders of the two boxes.

A few examples that illustrate the performance of the detection and segmentation methods described above are presented in Figures 3-6. In Figure 3, the real targets are encapsulated in rectangular boxes.



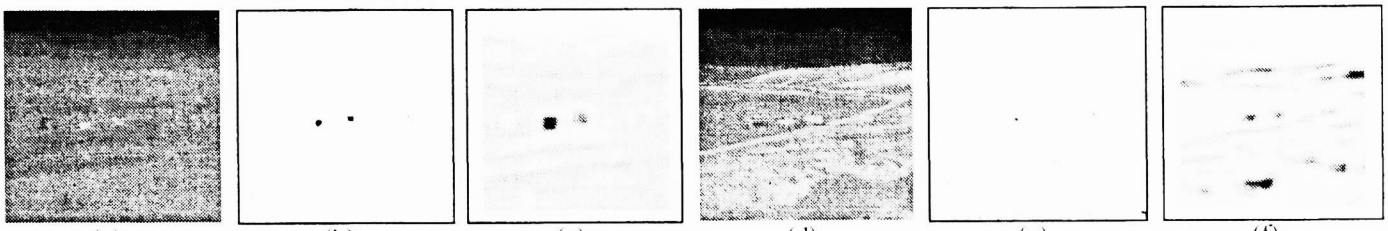
(a) (b) (c) (d) (e) (f)

Figure 3. Edge-based confusing form detection and segmentation: (a) and (d) originals, (b) and (e) edge maps, (c) and (f) segmented objects



(a) (b) (c) (d)

Figure 4. Scale-space smoothing: (a) and (c) originals, (b) and (d) smoothed images.



(a) (b) (c) (d) (e) (f)

Figure 5. Comparative detection by hit-miss transform and window tracker: (a), (d) originals, (b), (e) detection by the hit-miss transform, (c), (f) detection by the window tracker.

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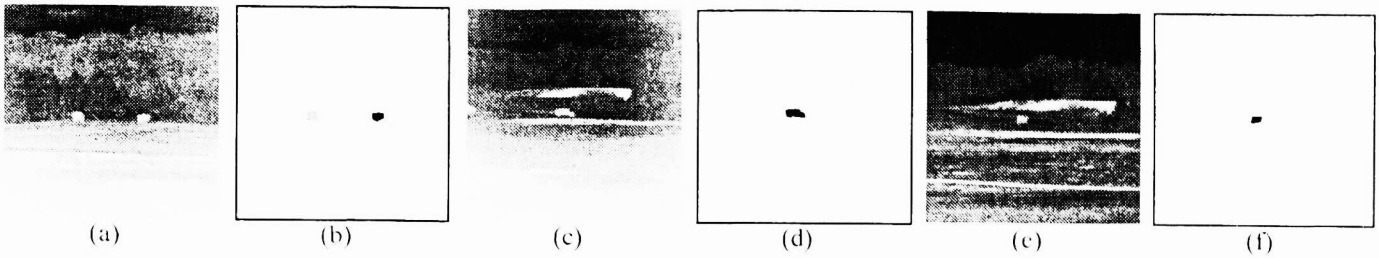

Figure 6. Comparative detection by hit-miss transform and window tracker: (a), (c), (e) original images, (b), (d), and (f) segmented targets.

Further investigations are being conducted that are aimed at developing a composite detection and segmentation method.

5. SYSTEM ARCHITECTURE

The general architecture of the adaptive modeling system is illustrated in Figure 7. Based on prior information about the range at which the images were acquired, a switch will activate the global or the local clutter modeling subsystem. Each of these subsystems consists of two concurrent processes: the observer test and the feature extraction process. In the global clutter modeling approach, the observer test provides the system with image scores that measure the intrinsic complexity of the respective images in terms of target detection. An image score can be given by the target detection probability for that image calculated over the entire set of subjects involved in the test. In parallel, an automated process extracts from the scenes the global features that will be put in correspondence with their respective scores to form the training data set for the evolutionary learning process.

If relatively small targets are expected in the scene, the switch will invoke the local clutter modeling subsystem. Essentially, its architecture is similar with the one employed for global clutter modeling but contains two additional modules, namely, the confusing form segmentation and its prerequisite, the preprocessing module. In the local observer test, performance scores are assigned to each of the detected confusing forms. In the simplest case, these scores will be binary and will advise whether the respective form was considered a target or not by the observers tested.

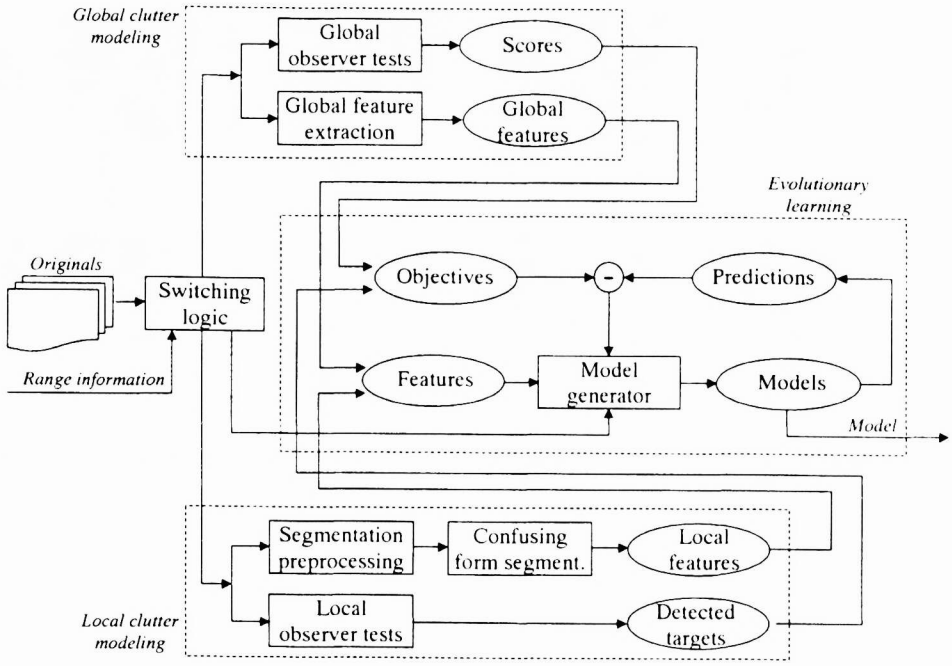

Figure 7. Architecture of the adaptive clutter modeling system.

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The central block of the architecture from Figure 7 is the evolutionary learning module, therein represented as a feedback process. The learning module receives objectives (scores) and features from both subsystems. The model generator block issues a model; this model generates predictions that are compared with the objectives. The cumulated error between the predictions and the objectives is fed back into the model generator that assigns it to the respective model as a fitness value. The evolutionary process will be driven by fitness and will evolve increasingly accurate models while seeking for a global minimum onto the fitness hypersurface.

6. PRELIMINARY RESULTS

In order to examine the descriptive power of the local features defined in the previous sections, a simple observer performance test was conducted that employed a 3-second stare time. The goal of this experiment was to categorize a set of 30 images into two detection-related classes (difficult and easy). Figure 8 depicts the 30 images employed in this test. The imagery contains various types of vehicles (M60, M1, M2-Bradley, T72, M113, and HMMWV) that are identified in the captions below each frame. Several clutter elements are also present in the selected scenes, such as sensor calibration boards, rocks, trees, and non-target military equipment (tents).

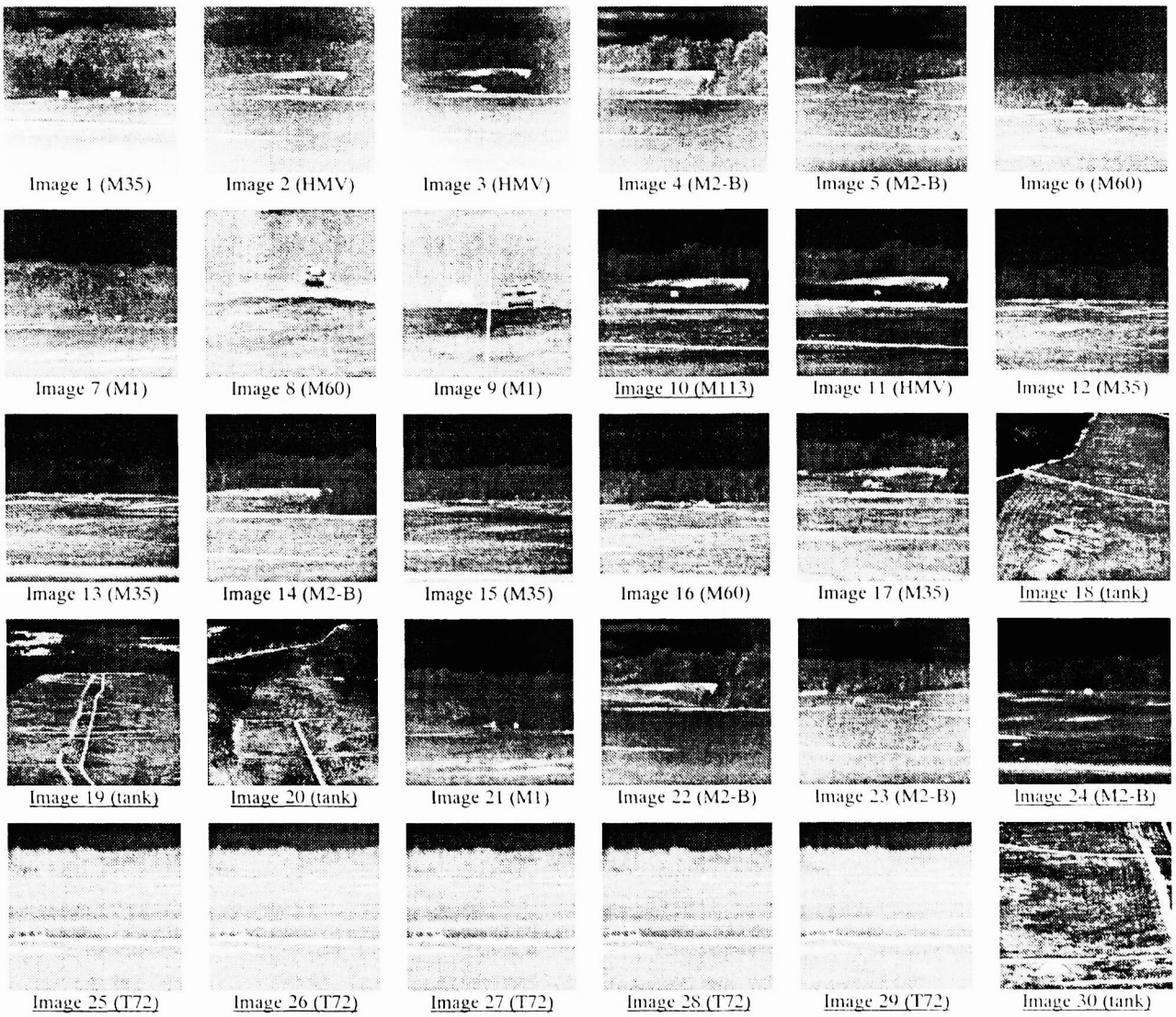

Figure 8. Images employed in the observer test. The underlined titles denote images deemed difficult by the observer test (in which the targets were detected with less that 50% probability).

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A first look at the images employed in the test leads to the conclusion that the observers were inconsistent. Perceptually similar images in terms of complexity have been labeled differently, such as images 10 and 11. The test was not conducted under laboratory conditions, no training was scheduled, and the subjects were not qualified for this type of problem. Nevertheless, these inconsistencies have been useful in evaluating the robustness in real case conditions of our modeling approach.

Three of the top-scoring models are shown in Table 4. They duplicated with 93.3% accuracy (2 misclassified images out of 30) the conclusions generated by the test. Each of the models employed only two global features; among them, a consistent presence is *Area* (the area of the target expressed in pixel squared).

Model program	Model expression	Misclassified images

`[ 1] Class = 0.853967;`
`[ 2] Class = Class - Area / 2580;`
`[ 3] Class = Class - Area / 2580;`
`[ 4] Class = Class / (I/0.7215);`
`[ 5] Class = 1 - Class;`
`[ 6] if (Class <= 0.0) Class = 0.0;`
`else Class = 1.0;`
	$Class = \begin{cases} "difficult", if\ 1+0.7215I\left(0.853967-\frac{Area}{1290}\right) \geq 0, \\ "easy", otherwise \end{cases}$	• Image 11 (rated difficult)
		• Image 12 (rated difficult)

`[ 1] FUZZY = NOT(Area);`
`[ 2] Class = TC;`
`[ 3] Class = Class - FUZZY;`
`[ 4] Class = Class / 0.169418;`
`[ 5] Class = Class - 0.169418;`
`[ 6] if (Class <= 0.0) Class = 0.0;`
`else Class = 1.0;`
	$Class = \begin{cases} "difficult", if\ -0.1694+5.90256\left(-1+\frac{Area}{2580}+1.6014TC\right) \geq 0, \\ "easy", otherwise \end{cases}$	• Image 10 (rated easy)
		• Image 24 (rated easy)

`[ 1] Class = Area / 2580;`
`[ 2] Class = Class / 0.0637347;`
`[ 3] Class = Class - POE / 0.09748;`
`[ 4] if (Class <= 0.0) Class = 0.0;`
`else Class = 1.0;`
	$Class = \begin{cases} "difficult", if\ -0.060814\,Area-10.2585\,POE \geq 0, \\ "easy", otherwise \end{cases}$	• Image 10 (rated easy)
		• Image 24 (rated easy)

Table 4. Top-scoring models produced after the binary classification experiment.

It is interesting to note that an underlying pattern may be identified behind the images misclassified by these models. It has been noticed in the previous paragraphs that the subjects tended to miss the target from Image 10, although the location of that target is quite obvious to the casual observer. The generated models did not categorize Image 10 as difficult. Indeed, the first model attempted to accommodate the "difficult" rating of Image 10 by considering two other similar images, Image 11 and Image 12, as "difficult". On the other hand, the second and the third model could not accommodate this rating for Image 10 and misclassified Image 24 as well. These disagreements might be explained by the poor training of the subjects.

In order to produce reliable and robust models, our system will have to train extensively, on a large palette of images. New experiments are scheduled for the near future that will address the reliability of the local features.

7. CONCLUSIONS

This study reports on the principles and the preliminary results of an ongoing research aimed at designing an adaptive modeling methodology for predicting the target detection performance with infrared imagery. The system is centered on a learning method powered by an evolutionary programming engine. Essentially, the learning method captures the relations between a set of objective features extracted from a collection of images and the observed performance of a human subject or a seeker emulator on the same collection. An error-driven, iterative process conducted through genetic operators derives these relations in an analytical form suitable for modeling. The features are divided into local and global features based on the extent of their scope; local features describe the degree of discrimination between the target and its local background, whereas global features quantify the amount of clutter present in the entire scene. Consequently, local features are applied for scenes with small targets, as opposed to the global features, which apply for short-range images. An additional component of the local clutter modeling is the confusing form detection and segmentation module, concerned with finding all the objects

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within a scene that resemble a target. It is within neighborhoods of these objects that the local features are calculated. The preliminary results reported are based on an observer test that involved 30 images and 8 subjects. A model was developed that performed a binary classification of the imagery (in high-clutter versus low-clutter images) with an accuracy of 93.3%.

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